

**SECTION A**Answer **all** questions.

1. A hydrocarbon is either cyclopentane or pent-1-ene.

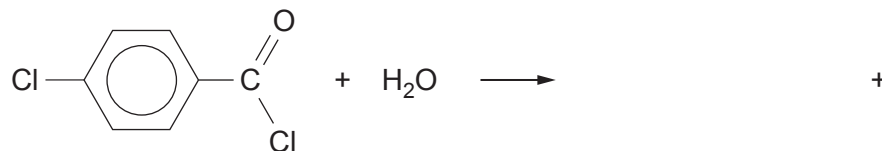
Use the **Data Booklet** to choose a characteristic infrared absorption that will positively identify the compound as pent-1-ene. [1]

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2. Give the **name** of a compound of formula  $C_4H_8$  that will decolourise acidified potassium manganate(VII). [1]

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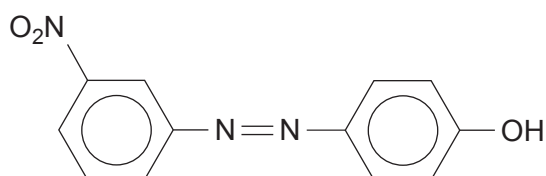
3. Complete the equation for the reaction shown below. [1]



4. Give the structure of a compound of molecular formula  $C_4H_8Br_2$  whose low resolution  $^1H$ NMR spectrum consists of three peaks. [1]



5. The formula of an azo dye, produced by reacting a diazonium compound with phenol, is shown below.



Write the formula of the amine that is used to produce this diazonium compound.

[1]

6. Ethanal reacts with hydrogen cyanide to give 2-hydroxypropanenitrile,  $\text{CH}_3\text{CH}(\text{OH})(\text{CN})$ .

(a) State the type of mechanism occurring during this reaction.

[1]

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(b) Draw the structures of the two enantiomers of 2-hydroxypropanenitrile.

[1]



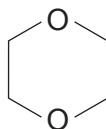
7. Three isomers have the molecular formula  $C_4H_8O_2$ .

(a) (i) Give the structure of an isomer that gives bubbles of carbon dioxide when tested with aqueous sodium hydrogencarbonate. [1]

(ii) Give the structure of an isomer that is both a ketone and a secondary alcohol. [1]

(b) Describe and explain the high resolution  $^1H$ NMR spectrum of 1,4-dioxane.

The position of the signal(s) is **not** required. [1]



1,4-dioxane

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